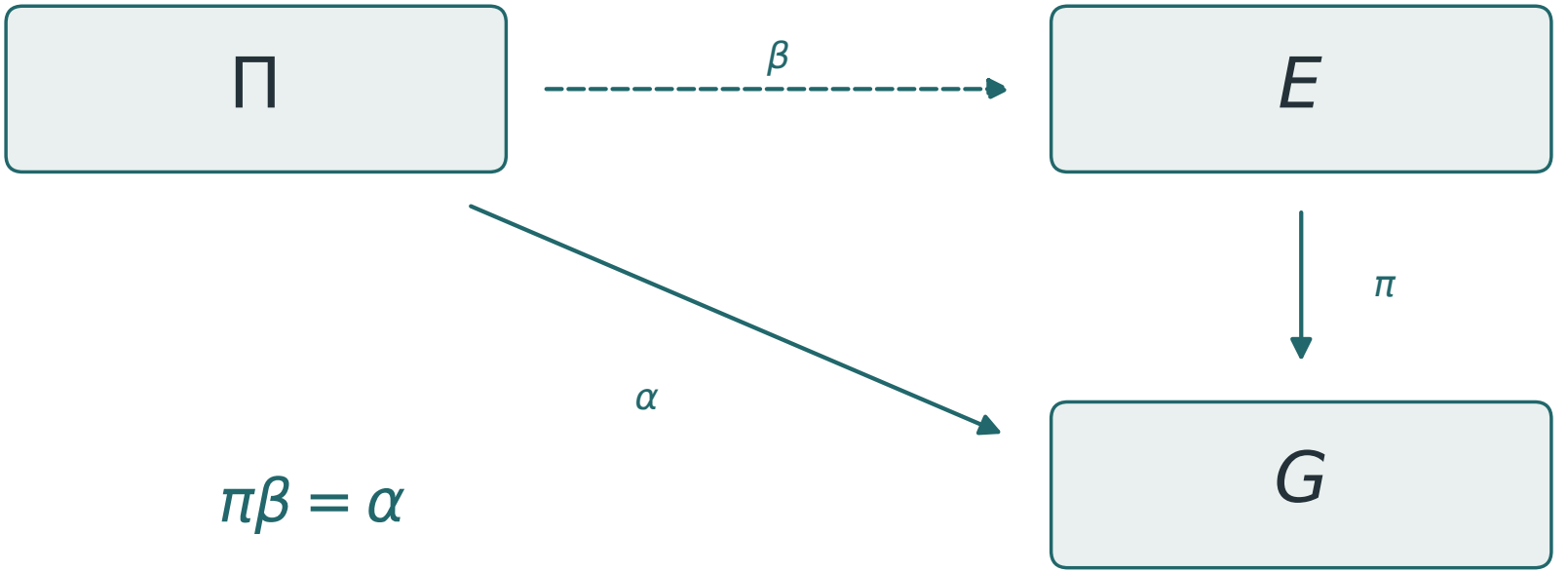
An embedding problem asks for a compatible lift

MODEL / the domain and surjectivity requirements matter



Weak solution: a continuous lift. Proper solution: the lift is surjective.

Finite toy: $\Pi = C_2$, $E = C_4$, $G = C_2$, $\alpha = \mathrm{id}$ has no lift through parity.